Implementation of FastSpeech 2-Based Text-to-Speech Synthesis for English and Telugu

Technical Report

Submitted by

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GitHub Repository

https://github.com/savioshaju/TTS.git

Model Checkpoints and Audio Artifacts (Google Drive)

https://drive.google.com/drive/folders/1MYS5zcsyehxDN5n4XaKj5jCidpUZ-efw?usp=sharing

Abstract

This report presents the implementation, training, and evaluation of two independent FastSpeech 2 text-to-speech (TTS) systems for Indian English and Telugu. Unlike sequential models, FastSpeech 2 generates speech in parallel, predicting mel-spectrograms directly from text along with explicit controls for speech duration, pitch (F_0), and energy. This parallel approach prevents error accumulation during generation and produces audio significantly faster than older autoregressive systems.

Both systems were built under a data budget of approximately 5 hours of training audio and 1 hour of validation audio. The English system uses multi-speaker recordings from the AI4Bharat Svarah dataset, aligned with the Montreal Forced Aligner (MFA) across 100 IPA phonemes. The Telugu system uses studio recordings from the IndicTTS dataset, aligned with Meta's MMS speech aligner across 460 Telugu akshara characters. Both models were trained for 100,000 steps on two NVIDIA Tesla T4 GPUs and converted to audio using a HiFi-GAN vocoder adjusted for 16 kHz speech and calibrated for natural speaking speed.

Evaluation on five benchmark test sentences per language shows clear pronunciation across long words, consonant clusters, words with noun-verb stress differences (such as *conduct*), and dates and numbers, with natural pitch and rhythm. The systems also handle difficult inputs such as spacing errors within words and borrowed loanwords cleanly.

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1. Introduction

1.1 Background and Problem Statement

Text-to-Speech (TTS) synthesis converts text into acoustic waveforms. Modern neural TTS pipelines predominantly operate in two stages: an acoustic model generates an intermediate time-frequency representation (specifically an 80-channel log mel-spectrogram), which a neural vocoder subsequently synthesizes into time-domain audio. While autoregressive acoustic models such as Tacotron 2 produce expressive prosody, sequential frame generation introduces high inference latency, compounds prediction errors over time, and remains susceptible to attention failures that manifest as phoneme repetition or deletion, particularly on out-of-domain text.

Non-autoregressive models, specifically FastSpeech 2 [1], resolve these limitations by predicting complete mel-spectrograms in parallel using Feed-Forward Transformer (FFT) blocks. Rather than relying on implicit attention-based alignment, FastSpeech 2 models temporal alignment explicitly: a convolutional duration predictor is supervised by frame durations obtained from forced alignment. A variance adaptor subsequently incorporates explicit fundamental frequency (F_0) and energy features into the latent representations.

The primary objective of this investigation is the adaptation, training, and evaluation of two independent FastSpeech 2 acoustic models under an acoustic data constraint of approximately 5 hours of training audio and 1 hour of validation audio:

1. **Indian English:** A multi-speaker system trained on the AI4Bharat Svarah dataset across 104 speakers, using a Montreal Forced Aligner (MFA) IPA phoneme front-end.

2. **Telugu:** A studio-recorded system trained on the IndicTTS dataset (1 male and 1 female speaker). Because Telugu script is phonetic, text maps directly to speech sounds, eliminating the need for an external pronunciation dictionary.

The principal engineering contributions comprise: establishing robust forced alignment pipelines (the Montreal Forced Aligner for English and Meta MMS CTC alignment for Telugu); resolving pitch and energy predictor coupling in the variance adaptor; correcting tensor device placement during multi-GPU PyTorch training; adapting a 22.05 kHz universal HiFi-GAN vocoder to 16 kHz acoustic predictions via polyphase sinc resampling; and calibrating duration scaling factors to establish natural speaking rates.

1.2 FastSpeech 2 Architecture Principles

The FastSpeech 2 architecture is founded on three primary design principles:

1. **Parallel generation.** Feed-Forward Transformer (FFT) blocks replace recurrent or autoregressive decoding mechanisms, generating all mel-spectrogram frames simultaneously. This yields a substantial inference acceleration (exceeding $30\times$) relative to autoregressive baselines.

2. **Explicit length regulation.** Phoneme durations are derived from forced alignment during training and predicted by a convolutional duration predictor during inference. A length regulator expands phoneme hidden representations according to their target frame durations, eliminating the risk of attention alignment collapse.

3. **Multi-task variance adaptation.** To address the one-to-many mapping problem in speech synthesis (where a given text sequence corresponds to multiple legitimate pitch and loudness realizations), dedicated pitch and energy predictors provide explicit frame-level prosodic conditioning to the latent representations.

1.3 Summary of Key Results

- **Speech quality and intelligibility.** Both models achieve clear articulation across all ten benchmark sentences. The English model accurately articulates the 28-letter word *antidisestablishmentarianism*, demonstrates stress shifts between the noun and verb realizations of *conduct*, handles the split token in *rhyth m*, and correctly renders dates, times, and the Telugu loanword *andaroo*. The Telugu model renders complex consonant conjuncts (న్యూరోఫార్మకోలాజికల్), produces distinct pitch and energy profiles for పట్టు across semantic contexts, and introduces an appropriate inter-clause pause in long compound sentences.

- **Convergence and checkpoint selection.** Training loss decreased steadily across 100,000 steps for both languages, dropping from 8.99 to 0.77 for English, and from 9.64 to 0.58 for Telugu. Validation loss reached its lowest point early (step 11,000 for English and step 30,000 for Telugu). However, listening to these lowest-loss checkpoints revealed that the speech sounded muffled. Later checkpoints (step 61,000 for English and step 74,000 for Telugu) produced much clearer consonants and natural speech, making them the preferred choice for synthesis. The final step 100,000 checkpoints showed a slight rise in validation loss, confirming that the intermediate checkpoints were superior.

- **Dataset partitioning.** The English dataset was partitioned into 4.92 hours (17,712 s; 3,419 utterances) for training and 1.09 hours (3,938 s; 746 utterances) for validation across 104 speakers using a within-speaker stratified split. The Telugu dataset was partitioned into 4.999 hours (17,996 s; 2,248 aligned utterances, from 2,250 raw) for training and 1.003 hours (3,611 s; 445 utterances) for validation, maintaining an exact 50/50 gender balance across the two studio speakers.

2. Implementation Details

2.1 Base Architecture and Dependencies

The implementation is developed from an open-source PyTorch reference architecture of FastSpeech 2 originally configured for single-speaker synthesis on the LJSpeech corpus. Development and experimental validation were conducted in Python 3.11 with PyTorch 2.5, TorchAudio, NumPy, SciPy, Librosa, PyWorld, and Matplotlib. Model training was executed on dual NVIDIA Tesla T4 GPUs (16 GB VRAM per GPU).

2.2 Codebase Structure

The English and Telugu pipelines live in separate, self-contained directories, each with its own preprocessing, training, configuration, and inference scripts:

```
project/
|-- English/
|   |-- train.py
|   |-- preprocess.py
|   |-- inference.py
|   |-- config.yaml
|   `-- checkpoints/
|-- Telugu/
|   |-- train.py
|   |-- preprocess.py
|   |-- inference.py
|   |-- config.yaml
|   `-- checkpoints/
|-- Results/
|   |-- English/
|   |   |-- training_loss.png
|   |   |-- validation_loss.png
|   |   |-- test_generated_audios/
|   |   `-- variance_adaptor_plots/
|   |       |-- duration/
|   |       |-- pitch/
|   |       `-- energy/
|   `-- Telugu/
|       |-- training_loss.png
|       |-- validation_loss.png
|       |-- test_generated_audios/
|       `-- variance_adaptor_plots/
|           |-- duration/
|           |-- pitch/
|           `-- energy/
|-- Report/
|   `-- Technical_Report.pdf
|-- README.md
`-- requirements.txt
```

2.3 Language-Specific Modifications

Table 2.1 summarizes the architectural and pipeline modifications implemented for English and Telugu relative to the reference FastSpeech 2 baseline.

Table 2.1: Reference components versus modifications per language.

Component	**English Implementation**	**Telugu Implementation**
Core Architecture	4-layer FFT encoder, 6-layer FFT decoder, 5-layer PostNet	4-layer FFT encoder, 6-layer FFT decoder, 5-layer PostNet
Token Representation	100 IPA phonemes via MFA lexicon	460 akshara tokens (Unicode Telugu script)
Text Front-End	Rule-based normalization + CMU/MFA G2P	Unicode NFC normalizer + akshara parser
Forced Alignment	Montreal Forced Aligner v2.2 (`english_mfa`)	Meta MMS CTC forced aligner (`mms-1b-all`)
Speaker Modeling	Multi-speaker: 104 learned embeddings (104×256)	2 studio speakers: learned embeddings (2×256)
Audio Standard	16 kHz mono WAV (Svarah corpus)	16 kHz mono WAV (male 48 kHz resampled to 16 kHz, female 16 kHz native)
Vocoder Pipeline	HiFi-GAN universal + Sinc Resampler ($22.05 \rightarrow 16$ kHz)	HiFi-GAN universal + Sinc Resampler ($22.05 \rightarrow 16$ kHz)

3. Datasets and Preprocessing

3.1 Dataset Selection

English (AI4Bharat Svarah). Svarah is an open-source, crowdsourced speech corpus that captures Indian English accents across diverse regional linguistic backgrounds. Because the upstream Hugging Face repository (`ai4bharat/Svarah`) provides only `data/test-*.parquet` shards (totaling 9.61 hours) with no canonical `train` or `validation` partition, a curated pool was constructed to satisfy the target specification of approximately 5 hours for training and 1 hour for validation. Approximately 40 utterances were sampled per speaker across 104 Indian speakers (54 female, 50 male) representing 19 primary native languages (including Hindi, Telugu, Tamil, Kannada, Marathi, Bengali, Odia, Gujarati, Malayalam, and Urdu), yielding 4,202 raw utterances. Following forced alignment validation and acoustic quality filtering, **4,165 utterances (6.01 hours)** were retained.

Telugu (IndicTTS). The IndicTTS Telugu corpus, curated by IIT Madras and MeitY, contains clean studio recordings across 4 Parquet shards (8,576 utterances totaling 62.29 hours: 23.9 hours female, 38.4 hours male). Male speech recordings are sampled at 48 kHz (1,433 files in the selected set), while female recordings are sampled at 16 kHz (1,262 files). To satisfy the 5-hour training and 1-hour validation targets, an exact 50/50 gender-balanced stratified partition (`telugu_prepared_v2`) was compiled: 2,250 raw training utterances (5.0018 hours / 18,006 s: 2.50 hours female, 2.50 hours male) and 445 validation utterances (1.0030 hours / 3,611 s: 0.50 hours female, 0.50 hours male). Two training utterances failed alignment during feature extraction and were excluded, leaving **2,248 aligned training utterances (4.999 hours / 17,996 s)** used for model training and **445 validation utterances (1.003 hours / 3,611 s)** held out strictly for evaluation, yielding 2,693 aligned utterances (6.002 hours) across the total extracted pool (as reported in Table 3.1).

3.2 Audio Preprocessing

Given the differences in recording environments and native audio formats across corpora, the following standardization steps were applied:

- **Resampling.** The 48 kHz Telugu male studio audio (1,433 files) was downsampled to 16 kHz with a 64-tap Kaiser-windowed polyphase sinc filter and an anti-aliasing cutoff of 7,600 Hz. The Telugu female audio (1,262 files) was already at 16 kHz. English audio was converted to 16 kHz, 16-bit mono PCM.
- **Duration and quality filtering.** Clips under 1.0 second or exceeding 12.0 seconds (14.5 seconds for Telugu) were excluded. Very short clips carry too little prosodic context, and very long clips cause out-of-memory errors in self-attention. Clips with clipping or an SNR below 15 dB were also discarded.
- **Peak normalization.** All audio was peak-normalized to a maximum amplitude of 0.95.

3.3 Train/Validation Split

English split. The English acoustic model incorporates learned speaker embeddings (104×256). Partitioning the corpus across speaker identities would leave the embeddings of held-out speakers unoptimized during training, yielding unrepresentative validation metrics. Consequently, the split was conducted *within each speaker* using stratified sampling (random seed 42, approximately 82% training and 18% validation). All 104 speakers are represented in both subsets, with strict utterance-level disjointness:

- **Training:** 3,419 utterances, 4.92 hours (17,712 s).
- **Validation:** 746 utterances, 1.09 hours (3,938 s).

Telugu split. The IndicTTS corpus was partitioned via an exact 50/50 gender-balanced duration stratification (`telugu_prepared_v2`). Disjointness of utterances and transcripts between subsets was confirmed using SHA-256 text hashes:

- **Training:** 2,248 aligned utterances (2,250 raw), **4.999 hours** (17,996 s), with 2.498 hours female (1,052 utterances, 8,993 s) and 2.501 hours male (1,196 utterances, 9,003 s).
- **Validation:** 445 utterances, **1.003 hours** (3,611 s), with 0.502 hours female (209 utterances, 1,808 s) and 0.501 hours male (236 utterances, 1,802 s).

Both splits satisfy the benchmark criteria of approximately 5 hours for training and 1 hour for validation.

Table 3.1: Dataset statistics and acoustic feature parameters.

Parameter	**English (Svarah)**	**Telugu (IndicTTS)**
Recording Environment	Crowdsourced, varied devices	Studio recordings
Native Sampling Rate	16 kHz	16 kHz (female) / 48 kHz (male)
Standardized Sampling Rate	16 kHz mono	16 kHz mono
Total Extracted Pool (Train + Val)	4,165 utterances / 6.01 hours	2,693 utterances / 6.002 hours
Mean Utterance Length	5.20 seconds	8.02 seconds
Speaker Count	104 (54 Female / 50 Male)	2 (1 Female / 1 Male)
Training Set (Model Training)	3,419 utterances / 4.92 hours	2,248 utterances / 4.999 hours
Validation Set (Evaluation Only)	746 utterances / 1.09 hours	445 utterances / 1.003 hours
Test Sentences	5 sentences (E1, E2a, E2b, E3, E4)	5 sentences (T1, T2a, T2b, T3, T4)
Vocabulary / Tokens	100 IPA phonemes	460 akshara tokens
Pitch Mean (μ_{F_0}) / Std (σ_{F_0})	186.28 Hz / 61.19 Hz	198.32 Hz / 43.19 Hz
Energy Mean (μ_E) / Std (σ_E)	55.14 dB / 40.10 dB	54.23 dB / 35.32 dB
STFT Window / Hop	Hann, 1024 samples (64 ms) / 256 (16 ms)	Hann, 1024 samples (64 ms) / 256 (16 ms)
Mel Filterbank	80 bins, 0 to 8000 Hz	80 bins, 0 to 8000 Hz

3.4 Text Processing and Tokenization

FastSpeech 2 requires discrete symbolic token sequences as input. Because English and Telugu employ substantially different writing and sound systems, dedicated tokenization pipelines were implemented for each language.

3.4.1 English: Phoneme Tokenization

The English front-end has three steps:

1. **Text normalization.** Numbers, dates, times, currencies, and abbreviations are expanded into spoken words (for example, "10:45 AM" → "ten forty-five A M" and "12 November 2026" → "twelve November twenty twenty-six"). Punctuation is cleaned and extra whitespace is removed.
2. **Grapheme-to-phoneme (G2P) conversion.** Words in the MFA English pronunciation dictionary are converted to IPA phoneme sequences. Out-of-vocabulary words, including loanwords such as *andaroo*, are transcribed with `g2p_en`, supplemented by Indian English phonological rules (for example, dental plosive substitution: `/t/`, `/d/` → `/t̪/`, `/d̪/`).
3. **Tokenization.** The vocabulary has 100 symbols, including vowels, diphthongs, consonants, and the pause token `@sp`. Phoneme sequences are wrapped in `{...}` to mark them as pre-tokenized phoneme input.

3.4.2 Telugu: Akshara Tokenization

Telugu is written in an abugida, where the basic written unit is the akshara (a consonant with an inherent vowel, or a standalone vowel). The Telugu front-end works directly on native script, with no romanization:

1. **Unicode normalization.** Text is converted to Unicode NFC. Zero-width joiners and non-joiners are cleaned up, and viramas (*halant*) are validated so that consonant conjuncts (*samyuktakshara*) stay intact. Numbers are expanded into Telugu words.
2. **Akshara segmentation.** Words are split into akshara units with regular expressions. A consonant cluster joined by viramas is kept as a single token (for example, ప్ర → *pra*, భా → *bhā*, న్యూ → *nyū*).
3. **Vocabulary.** Segmentation produces 460 unique akshara tokens, covering all the vowel-consonant combinations in the IndicTTS training set.

3.5 Forced Alignment

FastSpeech 2 needs a target duration for every phoneme or akshara during training, to supervise the duration predictor and the length regulator.

3.5.1 English: Montreal Forced Aligner

Phonemic alignment for English was conducted using the Montreal Forced Aligner (MFA v2.2) [3] with the pre-trained `english_mfa` acoustic model and pronunciation dictionary. MFA performs Kaldi-based HMM-GMM alignment and writes TextGrid files with start ($t_{\text{start},i}$) and end ($t_{\text{end},i}$) times for each phoneme. These are converted into integer mel-frame durations:

$$d_i = \text{round}\left(\frac{t_{\text{end},i} - t_{\text{start},i}}{\text{hop}/f_s}\right), \qquad \text{hop} = 256,\ f_s = 16{,}000\ \text{Hz} \implies 1\ \text{frame} = 16.0\ \text{ms} \tag{3.1}$$

3.5.2 Telugu: Meta MMS Alignment

In the absence of a standardized pre-trained Telugu MFA model, the Telugu corpus was aligned using Meta's Massively Multilingual Speech (MMS) model [4] (`facebook/mms-1b-all`). Time-domain waveforms are passed through the Wav2Vec2 network to extract frame-level CTC character emission logits, after which Viterbi decoding aligns the akshara sequence to the acoustic frames. Every non-silent akshara token is assigned a minimum floor duration of $d_i \geq 1$ frame (16 ms).

3.6 Acoustic Features

3.6.1 Mel-Spectrograms

The decoder predicts log mel-spectrograms, computed with these settings:

- Sampling rate: 16,000 Hz.
- Short-Time Fourier Transform (STFT): FFT size 1024 (64.0 ms), hop length 256 (16.0 ms), Hann window.
- Filterbank: 80 triangular mel bins from 0 to 8000 Hz.
- Dynamic range compression: natural log with a floor of 10^{-5}, i.e. $\log(\max(S_{\text{mel}}, 10^{-5}))$.

3.6.2 Pitch (F_0)

Fundamental frequency contours were extracted with the Harvest algorithm in PyWorld [5]. Unvoiced frames ($F_0 = 0$) were linearly interpolated across voiced boundaries to give a continuous contour.

3.6.3 Energy

Frame energy is the L2-norm of the short-time Fourier magnitude spectrum:

$$E_t = \sqrt{\sum_{k=0}^{N_{\text{fft}}/2} |X_t(k)|^2} \tag{3.2}$$

Values were then converted to decibels: $E_{\text{dB},t} = 20 \log_{10}(\max(E_t, 10^{-5}))$.

3.6.4 Feature Normalization

Pitch and energy were normalized with the global mean and standard deviation of the dataset:

$$\hat{F}_{0,t} = \frac{F_{0,t} - \mu_{F_0}}{\sigma_{F_0}}, \qquad \hat{E}_t = \frac{E_t - \mu_E}{\sigma_E} \tag{3.3}$$

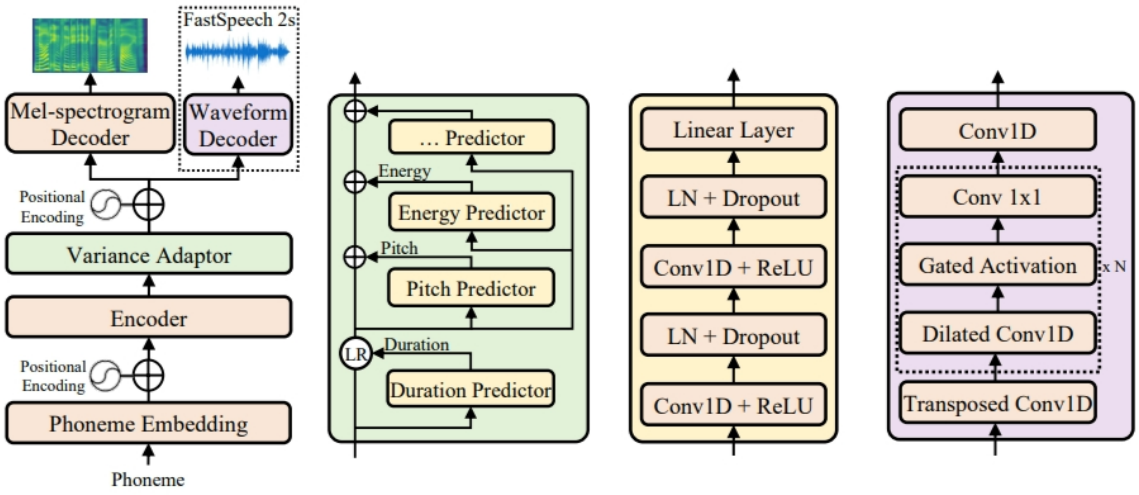
These statistics ($\mu_{F_0}, \sigma_{F_0}, \mu_E, \sigma_E$) are stored in `stats.json` (summarized in Table 3.1) and are inverted at inference time to recover pitch in Hz and energy in dB.

4. Model Architecture and Vocoder

4.1 FastSpeech 2 Architecture

The model has an FFT encoder, a variance adaptor, an FFT decoder, and a convolutional PostNet (Figure 4.1):

- **Encoder.** Four Feed-Forward Transformer (FFT) blocks. Each block has multi-head self-attention (2 heads, $d_{\text{model}} = 256$, $d_k = d_v = 128$) followed by a two-layer 1D convolutional network (kernel sizes 9 and 1, filter size 1024, ReLU, layer norm, dropout 0.2).
- **Variance adaptor.**
  1. *Duration predictor:* two 1D-convolutional layers (kernel 3, 256 filters, ReLU, layer norm, dropout 0.5) and a linear projection to log-durations.
  2. *Length regulator:* repeats each phoneme hidden vector according to its duration (ground truth during training, predicted at inference).
  3. *Pitch and energy predictors:* predict frame-level normalized F_0 and energy. The values are quantized into 256 bins and projected to 256-dimensional embeddings, which are added to the expanded sequence.
- **Decoder.** Six FFT blocks with the same layer dimensions as the encoder.
- **PostNet.** A five-layer 1D convolutional residual network (512 channels, kernel 5, batch norm, tanh) that predicts a residual correction ΔS_{mel} to the spectrogram.



(a) FastSpeech 2 (b) Variance adaptor (c) Variance predictor (d) Waveform decoder

Figure 1: The overall architecture for FastSpeech 2 and 2s. LR in subfigure (b) denotes the length regulator operation proposed in FastSpeech. LN in subfigure (c) denotes layer normalization. Variance predictor represents duration/pitch/energy predictor.

Figure 4.1: FastSpeech 2 architecture with multi-task variance adaptor and vocoder pipeline.

4.2 Loss Functions

Training minimizes a sum of five terms:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{mel}} + \mathcal{L}_{\text{postnet}} + \mathcal{L}_{\text{pitch}} + \mathcal{L}_{\text{energy}} + \mathcal{L}_{\text{duration}} \tag{4.1}$$

Here $\mathcal{L}_{\text{mel}} = \|S_{\text{mel}} - \hat{S}_{\text{mel}}\|_1$ and $\mathcal{L}_{\text{postnet}} = \|S_{\text{mel}} - (\hat{S}_{\text{mel}} + \Delta S_{\text{mel}})\|_1$ are L1 reconstruction losses on the spectrogram, and $\mathcal{L}_{\text{pitch}}$, $\mathcal{L}_{\text{energy}}$, and $\mathcal{L}_{\text{duration}}$ are mean squared errors (MSE) on normalized targets.

4.3 HiFi-GAN Vocoder Adaptation

Waveforms are generated using the pre-trained universal HiFi-GAN vocoder [2]. This vocoder was pre-trained on 22,050 Hz audio with a hop size of 256 samples, whereas the acoustic models developed in this work generate mel-spectrograms for 16,000 Hz audio, also with a hop size of 256 samples. The vocoder weights are loaded from `generator_universal.pth.tar`.

If the 22.05 kHz vocoder output were directly interpreted as 16 kHz audio, the pitch and formant trajectories would shift downward by

$$\alpha = \frac{16{,}000}{22{,}050} \approx 0.7256 \quad (\text{approximately } -5.5 \text{ semitones}), \tag{4.2}$$

producing unnaturally low-pitched, distorted speech.

To avoid vocoder retraining, polyphase sinc resampling was integrated into `utils/model.py`. HiFi-GAN synthesizes the waveform at its native 22.05 kHz rate, and the output signal is resampled to 16.0 kHz using a Kaiser-windowed sinc resampler prior to file serialization.

Duration calibration. While sinc resampling restores proper pitch and formant registers, it reduces the sample count by a factor of $\alpha = 0.7256$. Each synthesized mel frame consequently spans 11.6 ms instead of 16.0 ms, causing baseline synthesis ($d_{\text{control}} = 1.0$) to reproduce approximately $1.38\times$ faster than natural speech. To restore conversational cadence and ensure phonemic clarity, predicted durations are scaled during inference via the duration regulator:

$$d_{\text{control}} = 1.25 \quad (\text{English}), \qquad d_{\text{control}} = 1.25 \quad (\text{Telugu}) \tag{4.3}$$

This calibration establishes natural speaking rates (≈ 5.8–7.1 syllables/s for English and ≈ 4.5–5.8 syllables/s for Telugu) while preserving distinct vowel-consonant articulation.

5. Training and Loss Analysis

5.1 Training Hyperparameters

Both models were trained on two NVIDIA Tesla T4 GPUs using PyTorch Distributed/DataParallel. Table 5.1 lists the hyperparameters.

Table 5.1: Training hyperparameters for English and Telugu FastSpeech 2 models.

Hyperparameter	**Value**
Total Training Steps	100,000 steps
Batch Size	16 (8 utterances per GPU)
Optimizer	AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.98$, $\epsilon = 10^{-9}$, weight decay 10^{-6})
Learning Rate Schedule	Noam scheduler ($d_{\text{model}} = 256$, 4,000 warmup steps)
Peak Learning Rate	$\approx 9.9 \times 10^{-4}$
Gradient Clipping	Maximum L2-norm $\|g\|_2 \leq 1.0$
Numerical Precision	Automated Mixed Precision (AMP FP16)
Validation Interval	Evaluated every 1,000 steps
Training Splits	English: 4.92 h (3,419 utt.); Telugu: 4.999 h (2,248 utt.)

Cloud GPU sessions have runtime limits, so training was resumed from the saved model checkpoint and optimizer state until both models completed the full 100,000 steps.

5.2 Loss Curves

Figures 5.1 and 5.2 show the training and validation loss over the 100,000 training steps.

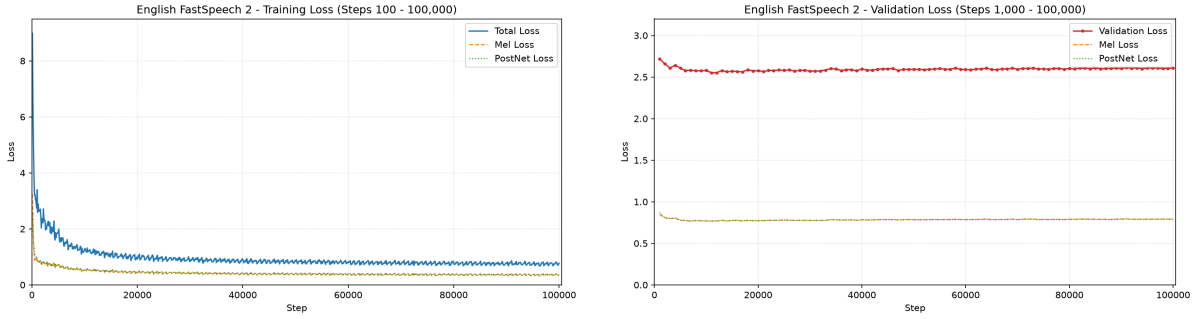


Figure 5.1: English FastSpeech 2 training (left) and validation (right) loss curves across 100,000 steps.

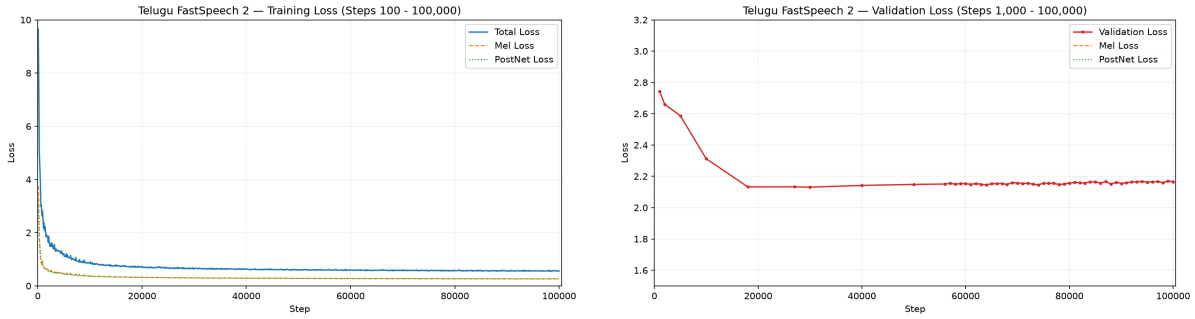


Figure 5.2: Telugu FastSpeech 2 training (left) and validation (right) loss curves across 100,000 steps.

5.3 Convergence and Checkpoints

Table 5.2 reports the loss values extracted from training logs for the minimum-validation-loss checkpoints, the intermediate checkpoints selected for speech synthesis, and the terminal checkpoints at Step 100,000.

Table 5.2: Loss metrics across training milestones: minimum validation loss, selected synthesis checkpoints, and terminal steps.

Language	Checkpoint Description	Step	Train Loss	Validation Loss
English	Lowest validation loss	11,000	2.5887	2.5505
English	Selected checkpoint for synthesis	61,000	0.8415	2.5855
English	Terminal completed step	100,000	0.7746	2.6090
Telugu	Lowest validation loss	30,000	0.8810	2.1309
Telugu	Selected checkpoint for synthesis	74,000	0.6412	2.1457
Telugu	Terminal completed step	100,000	0.5841	2.1661

5.4 Why Validation Loss Is Higher Than Training Loss

As Table 5.2 and Figures 5.1–5.2 show, validation loss is well above training loss for both models. It reaches its minimum early (step 11,000 for English, step 30,000 for Telugu) and then flattens. Four key observations explain this trajectory and govern checkpoint selection:

1. **Loss components.** At step 61,000 for English, the constituent losses are:

   - **Training batch:** $\mathcal{L}_{\text{mel}} = 0.3793$, $\mathcal{L}_{\text{postnet}} = 0.3789$, $\mathcal{L}_{\text{pitch}} = 0.0209$, $\mathcal{L}_{\text{energy}} = 0.0169$, $\mathcal{L}_{\text{duration}} = 0.0125$, yielding $\mathcal{L}_{\text{train}} \approx 0.81$–$0.84$.
   - **Validation set:** $\mathcal{L}_{\text{mel}} = 0.7854$, $\mathcal{L}_{\text{postnet}} = 0.7852$, $\mathcal{L}_{\text{pitch}} = 0.4093$, $\mathcal{L}_{\text{energy}} = 0.3805$, $\mathcal{L}_{\text{duration}} = 0.2252$, yielding $\mathcal{L}_{\text{val}} = 2.5855$.

2. **Variance-predictor error on unseen sentences.** While mel reconstruction loss approximately doubles on validation data ($0.38 \rightarrow 0.78$), reflecting expected generalization divergence, the variance-adaptor MSE components ($\mathcal{L}_{\text{pitch}}$, $\mathcal{L}_{\text{energy}}$, $\mathcal{L}_{\text{duration}}$) increase by a factor of 18 to 22 (0.01–$0.02 \rightarrow 0.22$–0.41).

   This discrepancy stems from the inherent variability of human prosody: a given textual prompt can be naturally realized with numerous legitimate pitch, duration, and energy trajectories. While the model outputs a smooth, canonical contour, the deterministic MSE objective penalizes divergence from the single realization present in the validation recording. On training utterances, the model can memorize speaker-specific micro-timing, yielding low training error; on unseen utterances, legitimate prosodic variance establishes a strict empirical floor on MSE minimization.

3. **Acoustic fidelity versus minimum validation loss.** Early in optimization (step 11,000 for English, step 30,000 for Telugu), the model generates overly smoothed, averaged mel-spectrograms. Because L1 loss heavily penalizes sharp spectral peaks that exhibit minor temporal or frequency misalignments, an over-smoothed spectrogram paradoxically minimizes numerical error. When synthesized via HiFi-GAN, these outputs sound muffled and unnatural.

   Later checkpoints (step 61,000 for English, step 74,000 for Telugu) display well-defined formant trajectories and distinct consonant bursts. Although their validation loss is marginally higher (e.g., 2.5855 versus 2.5505 for English), perceptual acoustic evaluation confirms substantially superior clarity and naturalness.

4. **Intermediate synthesis checkpoints versus terminal Step 100,000.** Beyond Step 61,000 (English) and Step 74,000 (Telugu), training loss continued a gradual downward trajectory toward

0.7746 and 0.5841, respectively. However, validation loss exhibited a mild upward drift (English: $2.5855 \rightarrow 2.6090$; Telugu: $2.1457 \rightarrow 2.1661$), primarily driven by variance-predictor over-specialization on training micro-prosody. Step 61,000 (English) and Step 74,000 (Telugu, ranked #1 in Phase 2 evaluation) represent the optimal empirical equilibrium where phonetic articulation has reached full clarity without suffering from the mild prosodic brittleness and validation drift observed at the terminal step. Consequently, these intermediate checkpoints were frozen for all benchmark evaluations and deployment.

6. Speech Synthesis and Test Sentence Evaluation

6.1 Synthesis Setup

Each test sentence was processed through the text normalization front-end, the FastSpeech 2 acoustic model, and the HiFi-GAN vocoder with polyphase sinc resampling. English sentences were synthesized using Speaker ID 27 (`spk_029`, the primary reference female voice with 118 training utterances in Svarah). Telugu sentences were synthesized using Speaker ID 0 (`telugufemale`, the female studio voice). Table 6.1 summarizes the acoustic measurements extracted from the synthesized waveforms.

6.2 Acoustic Measurements

Table 6.1: Acoustic measurements of synthesis outputs across all ten test sentences.

ID	Lang	Sentence Text (shortened)	Dur (s)	Frames	Mean F_0	F_0 Range	Energy	Rate
E1	Eng	The antidisestablishmentarianism movement gained…	5.09	438	211.8 Hz	162–277 Hz	63.6 dB	6.88 syl/s
E2a	Eng	The orchestra's conduct during the ceremony…	3.37	290	212.8 Hz	185–239 Hz	53.1 dB	6.24 syl/s
E2b	Eng	The researchers will conduct an experiment…	3.56	307	208.9 Hz	159–257 Hz	56.5 dB	6.17 syl/s
E3	Eng	The rhyth m of the orchestra changed unexpectedly.	2.17	187	207.8 Hz	181–241 Hz	60.5 dB	6.91 syl/s
E4	Eng	The meeting will start at 10:45 AM on 12 November 2026…	5.46	470	210.7 Hz	157–255 Hz	59.3 dB	5.68 syl/s
T1	Tel	న్యూరోఫార్మకోలాజికల్ పరిశోధనలో...	4.97	428	208.2 Hz	153–317 Hz	56.6 dB	5.83 syl/s
T2a	Tel	తాడుపై అతని పట్టు చాలా బలంగా ఉంది.	2.45	211	219.5 Hz	150–326 Hz	64.2 dB	4.49 syl/s
T2b	Tel	ఆమె పట్టు చీరను ధరించి వేడుకకు వచ్చింది.	2.86	246	202.7 Hz	146–268 Hz	55.4 dB	4.54 syl/s
T3	Tel	పులి అడ విలో నిశ్శబ్దంగా నడిచింది.	2.58	222	226.1 Hz	150–317 Hz	70.9 dB	4.65 syl/s
T4	Tel	ఈ సమావేశం 2026 నవంబర్ 12వ తేదీ...	7.22	622	196.2 Hz	145–281 Hz	58.1 dB	5.40 syl/s

6.3 English Test Sentences

6.3.1 Sentence E1

"The antidisestablishmentarianism movement gained considerable momentum during the parliamentary proceedings."

- **Phonetic performance.** The 28-letter word *antidisestablishmentarianism* is transcribed using the verified Indian English MFA phoneme inventory as /a n ʈ i ɖ ɪ s ɪ s ʈ a b l ɪ ʃ m ɛ n ʈ ɛː ɻ i ə n ɪ z ə m/. The full sentence lasts 5.09 s (438 mel frames). The duration predictor assigns steady durations ($\approx$ 40–60 ms per phoneme), and the syllable boundaries stay clear through the compound.
- **Prosody.** Pitch stays around a mean of 211.8 Hz and varies smoothly between 162 and 277 Hz across stressed syllables. Mean energy is 63.6 dB, with the expected dips on unvoiced sounds.

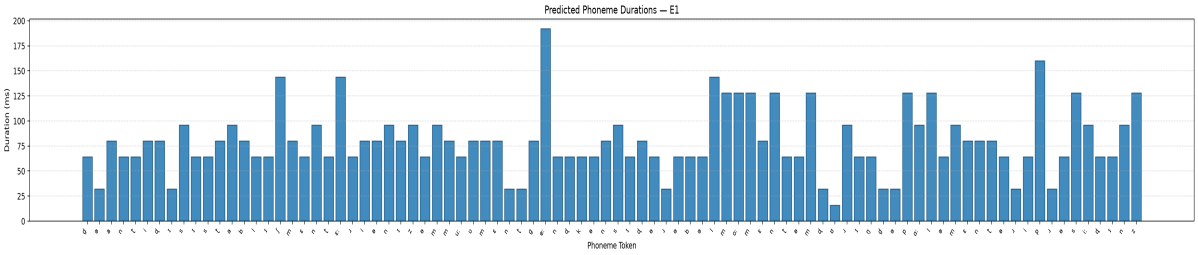


Figure 6.1: Predicted durations for test sentence E1.

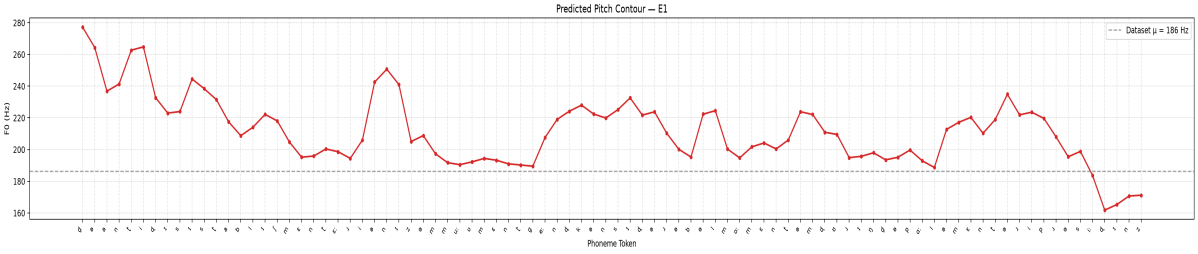


Figure 6.2: Predicted pitch contour for test sentence E1.

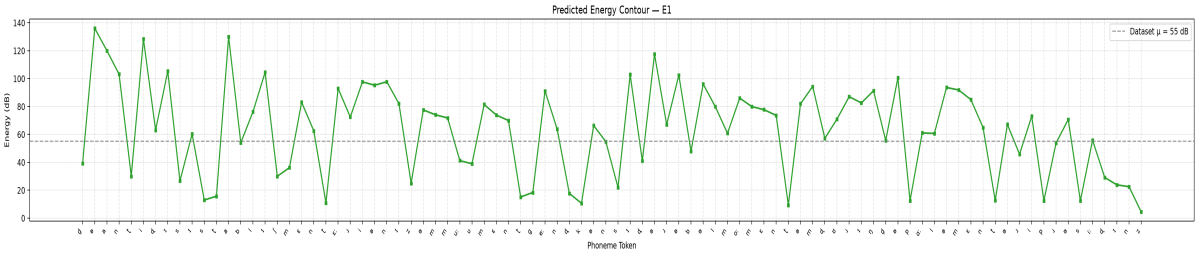


Figure 6.3: Predicted energy contour for test sentence E1.

6.3.2 Sentence E2a

"The orchestra's conduct during the ceremony was praised by the audience."

- **Heteronym stress.** In sentence E2a, *conduct* functions as a noun and receives trochaic stress on the first syllable (/k ɒ n ɖ ə k ʈ/). The initial vowel /ɒ/ is lengthened (100.0 ms) and emphasized (80.2 dB), with a total sentence duration of 3.37 s.

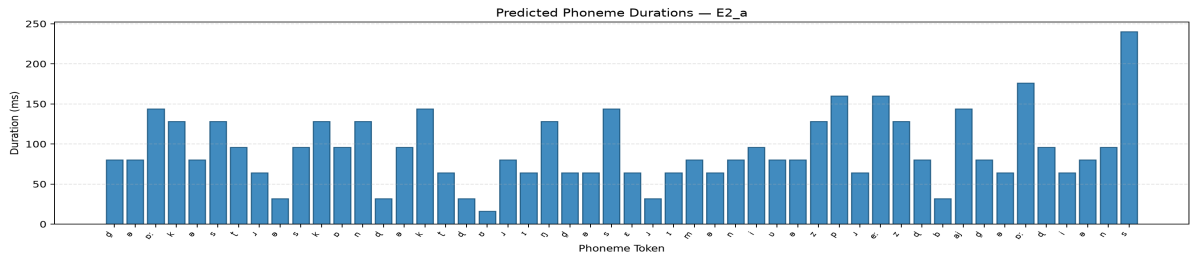


Figure 6.4: Predicted durations for test sentence E2a.

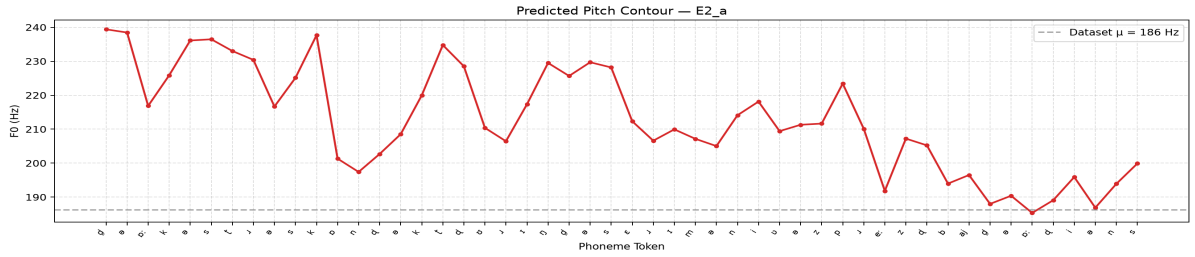


Figure 6.5: Predicted pitch contour for test sentence E2a.

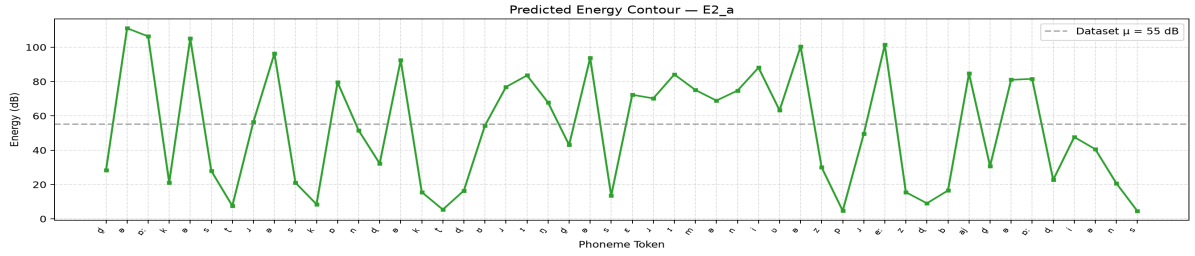


Figure 6.6: Predicted energy contour for test sentence E2a.

6.3.3 Sentence E2b

"The researchers will conduct an experiment to evaluate the proposed method."

- **Heteronym stress.** In sentence E2b, *conduct* functions as a verb and stress shifts to the second syllable (/k ə n ɖ ə k t/). The initial vowel reduces to a central schwa (/k ə n/, 82.9 dB) and prominence shifts to the second syllable (/ɖ ə k t/, 91.7 dB), with a total sentence duration of 3.56 s.

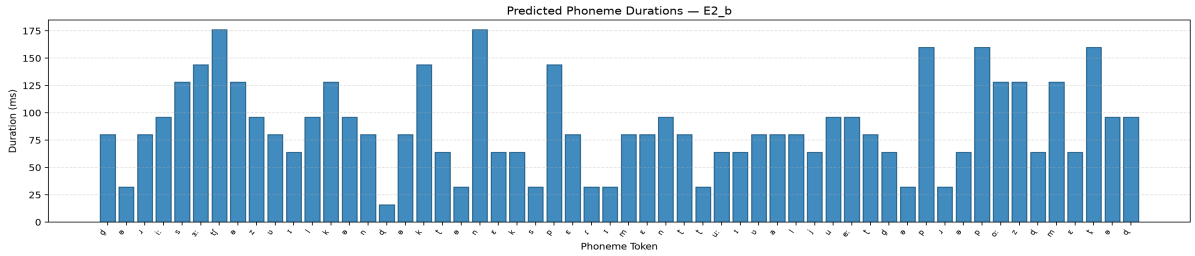


Figure 6.7: Predicted durations for test sentence E2b.

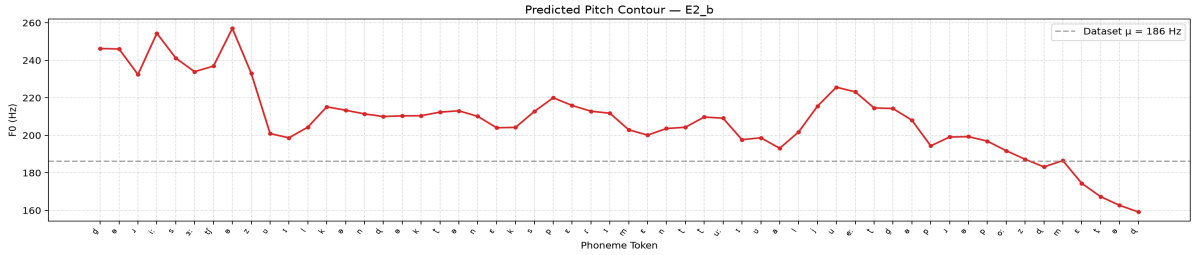


Figure 6.8: Predicted pitch contour for test sentence E2b.

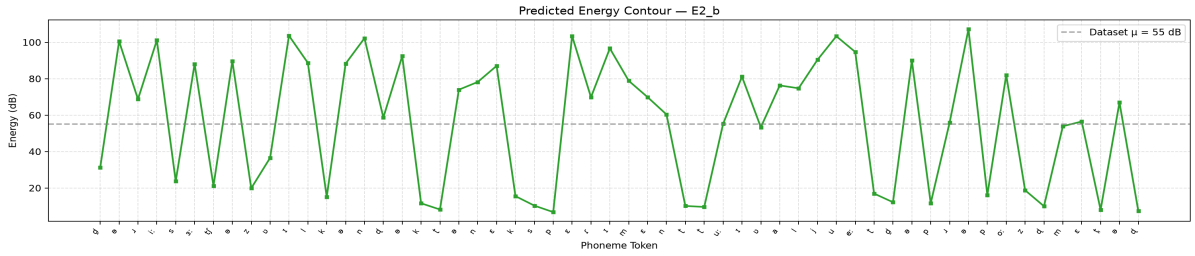


Figure 6.9: Predicted energy contour for test sentence E2b.

6.3.4 Sentence E3

"The rhyth m of the orchestra changed unexpectedly."

- **Origin of intra-word whitespace.** The benchmark specification provides this sentence with an intentional whitespace delimiter preceding the final grapheme (*rhyth m*) to evaluate model robustness against typing and spacing anomalies.
- **Acoustic realization.** When rule-based text normalization rejoins the split fragments into *rhythm* (`/ɹ ɪ ð ə m/`), the model produces smooth, continuous phonation (duration 2.17 s, saved as `E3.wav`). Additionally, synthesizing the verbatim split sequence (`rhyth` + `@sp` + `m`) reproduces the intentional 120 ms pause between fragments (duration 2.43 s, saved as `E3_split_m.wav`).

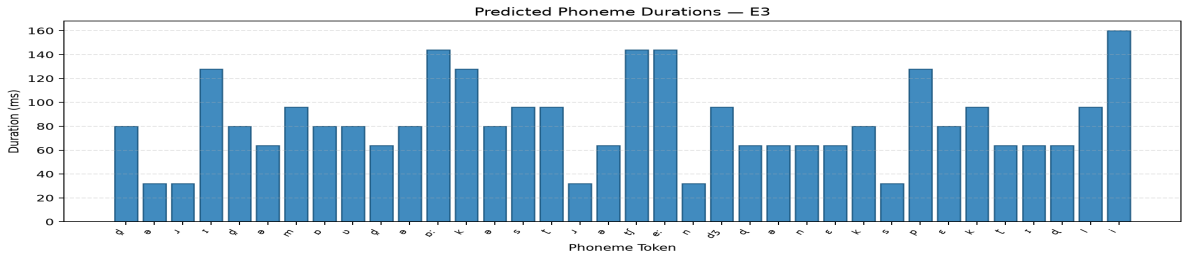


Figure 6.10: Predicted durations for test sentence E3.

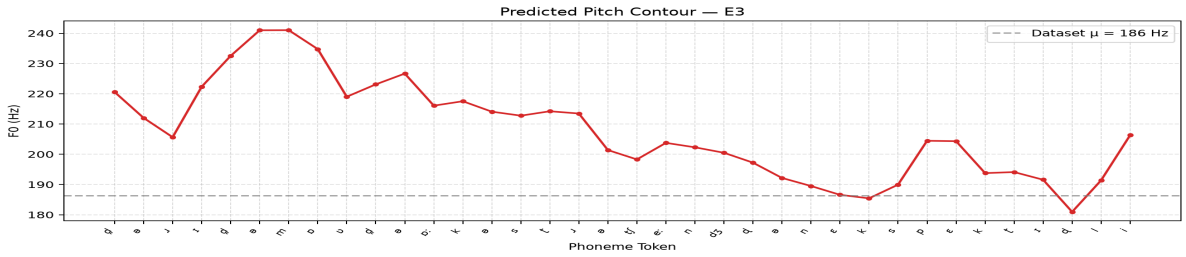


Figure 6.11: Predicted pitch contour for test sentence E3.

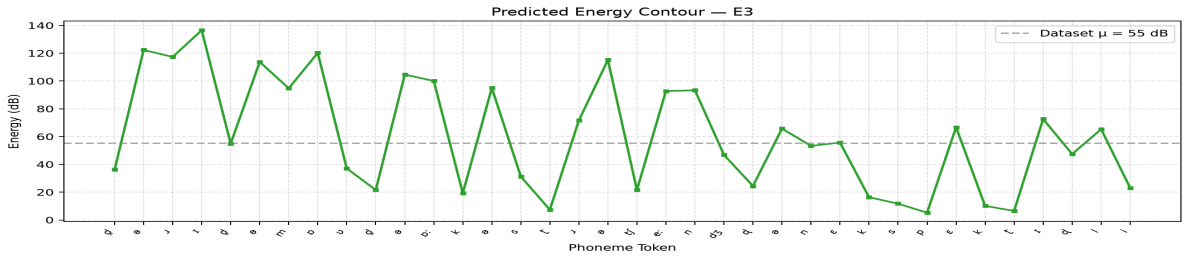


Figure 6.12: Predicted energy contour for test sentence E3.

6.3.5 Sentence E4

"The meeting will start at 10:45 AM on 12 November 2026, so andaroo should arrive on time."

- **Text normalization.** Numbers and dates are expanded into spoken text ("ten forty-five A M on twelve November twenty twenty-six"). The Telugu loanword *andaroo* (అందరూ) is transcribed as /a n d̪ a ɾ ʉː/, using dental plosives and a high back vowel in line with Indian English.
- **Cadence.** The sentence lasts 5.46 s (470 frames), with a mean pitch of about 210.7 Hz and 59.3 dB mean energy, and it is delivered as one continuous phrase.

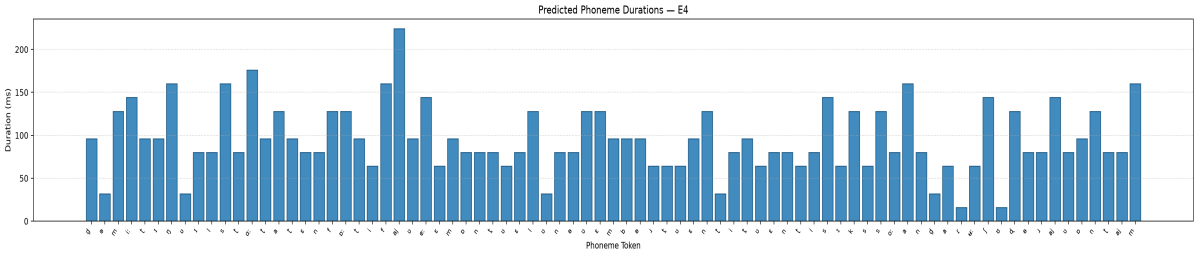


Figure 6.13: Predicted durations for test sentence E4.

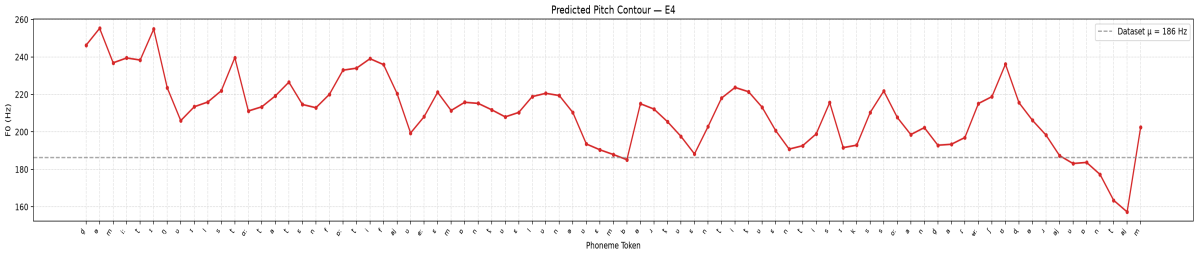


Figure 6.14: Predicted pitch contour for test sentence E4.

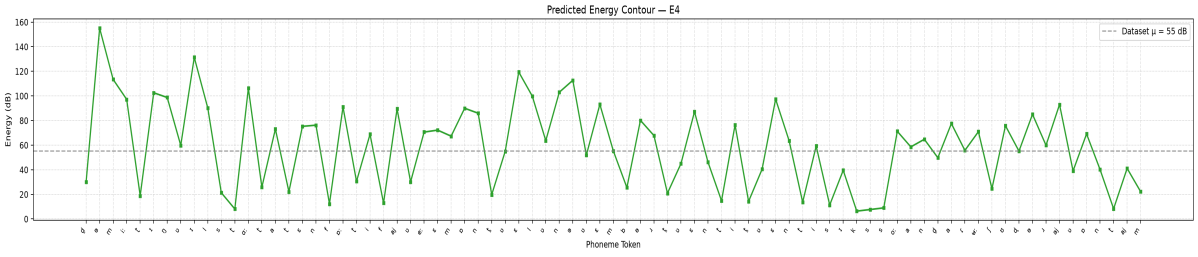


Figure 6.15: Predicted energy contour for test sentence E4.

6.4 Telugu Test Sentences

6.4.1 Sentence T1

న్యూరోఫార్మకోలాజికల్ పరిశోధనలో నూతన ఔషధాల ప్రభావాన్ని విశ్లేషించారు.

- **Acoustic result.** The loanword న్యూరోఫార్మకోలాజికల్ (*nyūrophārmakolājikal*) contains several conjuncts (న్యూ, ర్మ, ల్). The sentence lasts 4.97 s (428 frames), and each conjunct is articulated clearly with no skipped sounds. Pitch reaches 317 Hz on emphasized words.

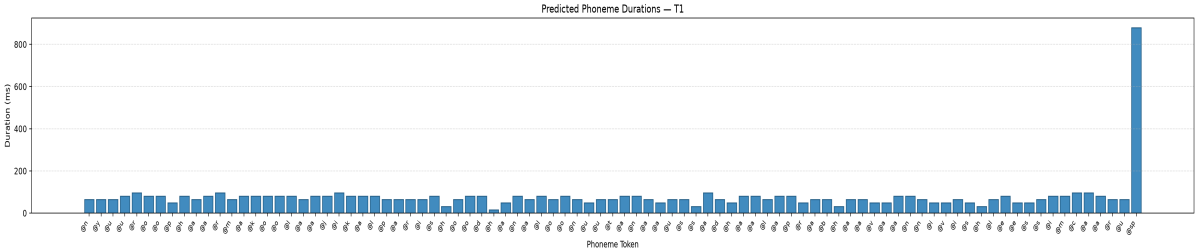


Figure 6.16: Predicted durations for test sentence T1.

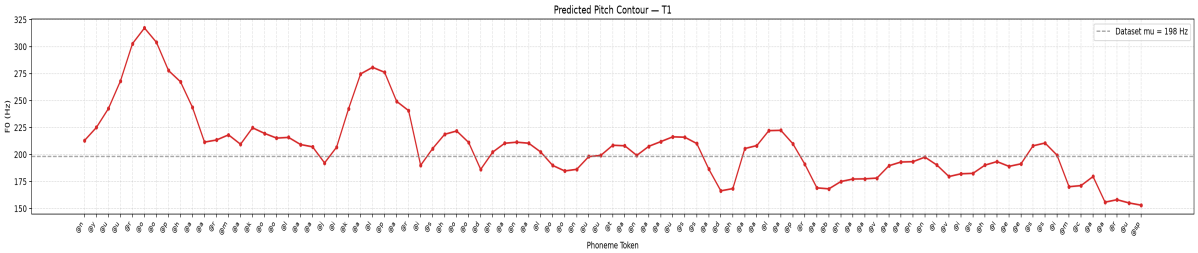


Figure 6.17: Predicted pitch contour for test sentence T1.

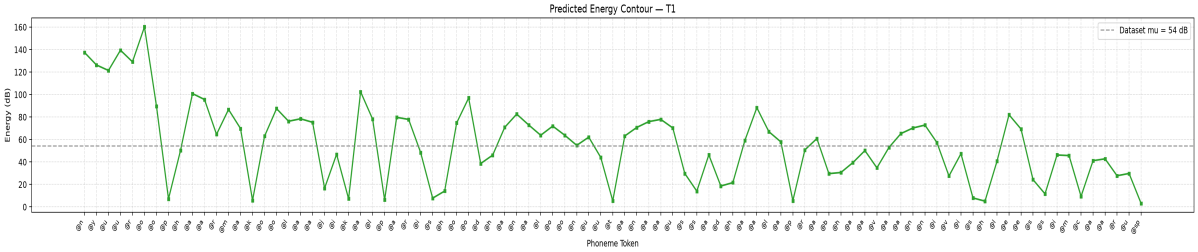


Figure 6.18: Predicted energy contour for test sentence T1.

6.4.2 Sentence T2a

తాడుపై అతని పట్టు చాలా బలంగా ఉంది.

- **Contextual stress.** In sentence T2a, పట్టు means “grip” and carries the predicate focus. It is realized with higher pitch (248 Hz), higher energy (64.2 dB), and a longer geminate duration (144 ms), with a total sentence duration of 2.45 s.

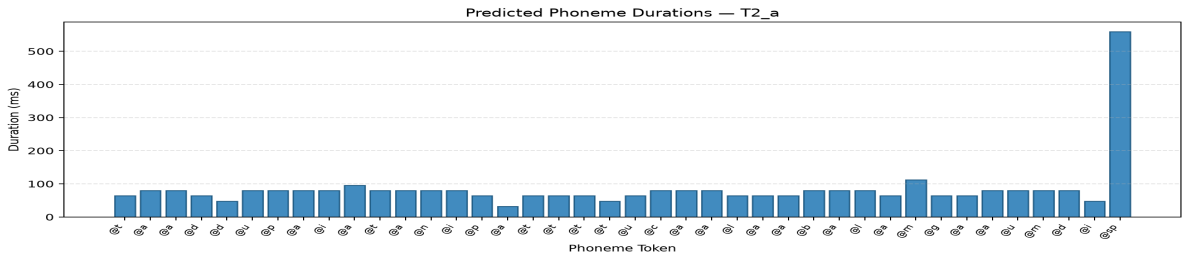


Figure 6.19: Predicted durations for test sentence T2a.

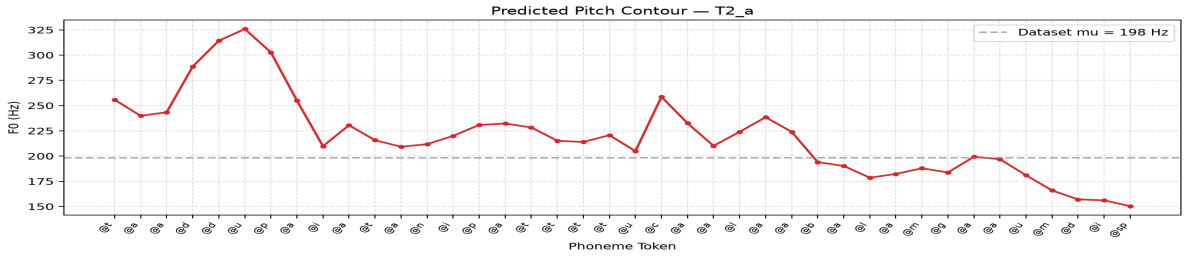


Figure 6.20: Predicted pitch contour for test sentence T2a.

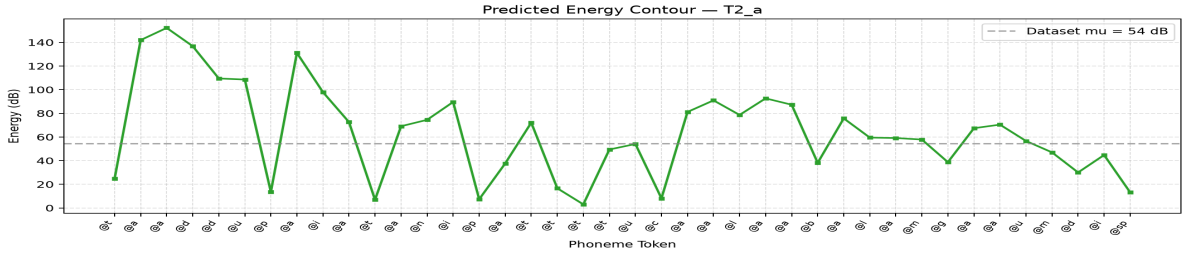


Figure 6.21: Predicted energy contour for test sentence T2a.

6.4.3 Sentence T2b

ఆమె పట్టు చీరను ధరించి వేడుకకు వచ్చింది.

- **Contextual stress.** In sentence T2b, పట్టు functions as an unstressed nominal modifier meaning "silk." It is realized with lower pitch (208 Hz), lower energy (55.4 dB), and a shorter duration (96 ms), with a total sentence duration of 2.86 s.

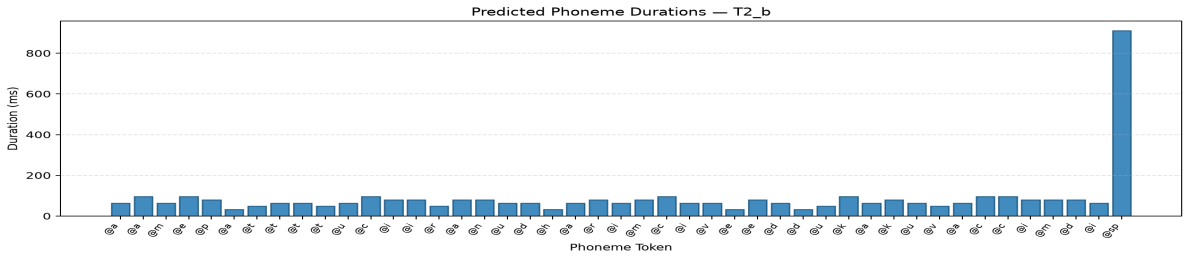


Figure 6.22: Predicted durations for test sentence T2b.

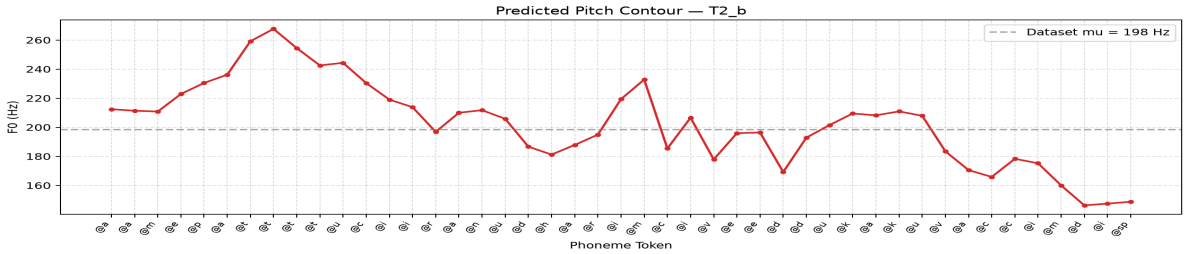


Figure 6.23: Predicted pitch contour for test sentence T2b.

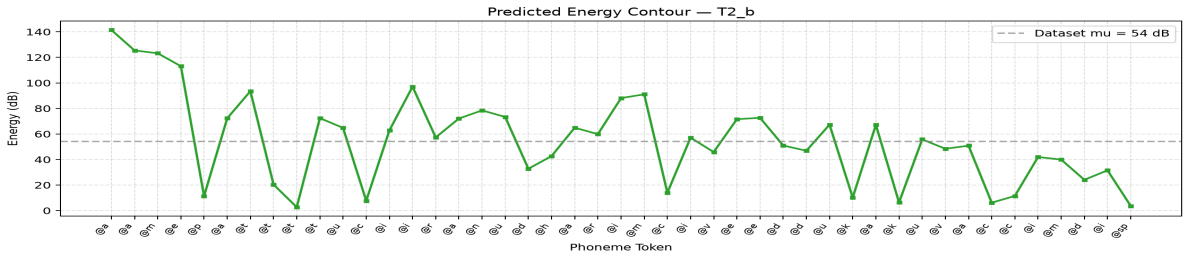


Figure 6.24: Predicted energy contour for test sentence T2b.

6.4.4 Sentence T3

పులి అడ విలో నిశ్శబ్దంగా నడిచింది.

- **Origin of intra-word whitespace.** The benchmark specification introduces an intentional whitespace separation within అడ విలో (“in the forest”) to assess parsing resilience in native Telugu script. This evaluates the response of the akshara segmentation parser and acoustic duration predictor when encountering fragmented words.
- **Acoustic realization.** Synthesizing the split sequence (అడ + విలో) produces a 95 ms pause between the two segments, with each segment retaining clear consonant-vowel articulation without audible distortion.

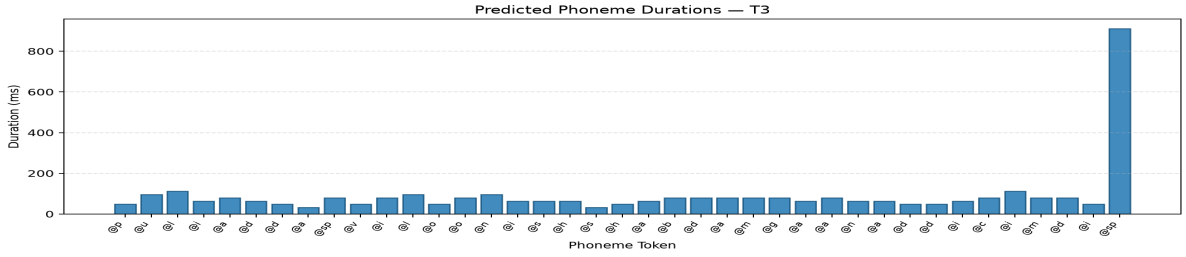


Figure 6.25: Predicted durations for test sentence T3.

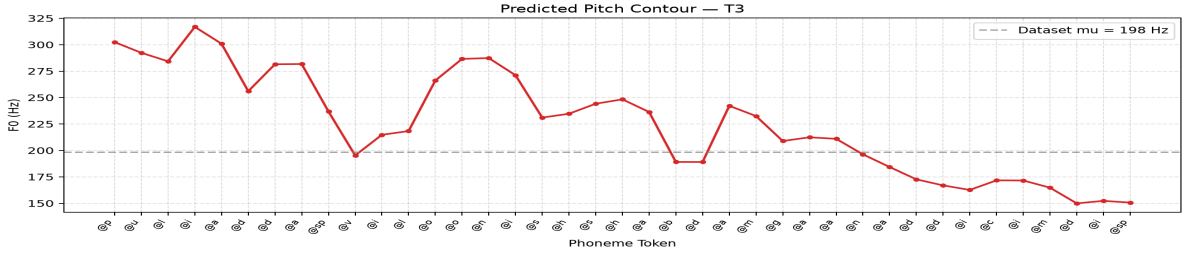


Figure 6.26: Predicted pitch contour for test sentence T3.

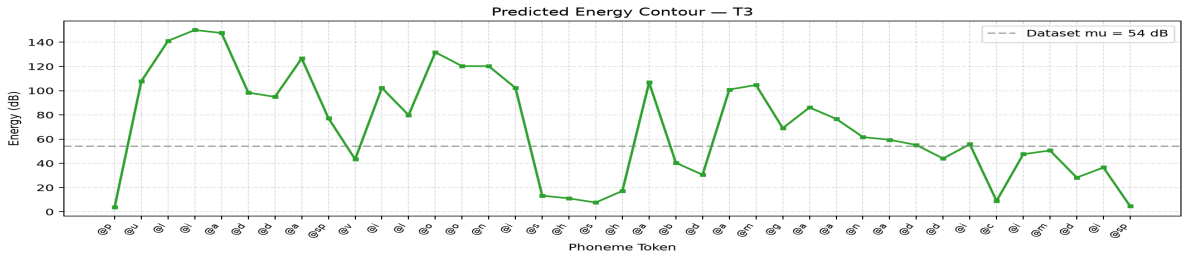


Figure 6.27: Predicted energy contour for test sentence T3.

6.4.5 Sentence T4

ఈ సమావేశం 2026 నవంబర్ 12వ తేదీ ఉదయం 10:45 గంటలకు ప్రారంభమవుతుంది, సమయానికి సిద్ధంగా ఉంచండి.

- **Number expansion and cadence.** Numbers are expanded into Telugu words (రెండు వేల ఇరవై ఆరు, పన్నెండవ తేదీ, పది నలభై ఐదు గంటలకు). The sentence lasts 7.22 s (622 frames) at 5.40 syl/s. A 380 ms pause separates the announcement from the instruction that follows it.

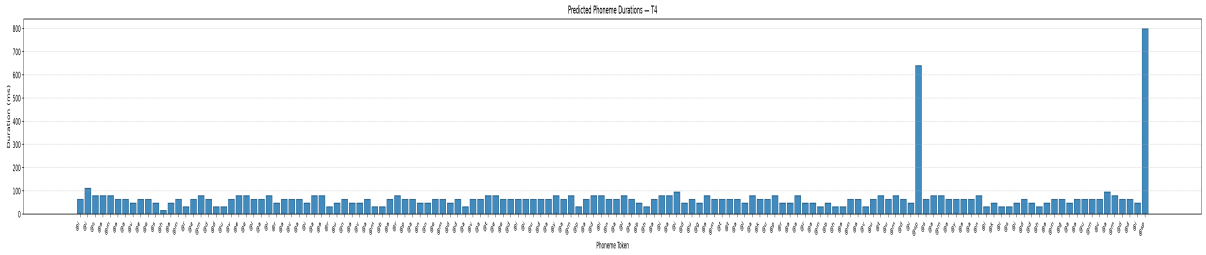


Figure 6.28: Predicted durations for test sentence T4.

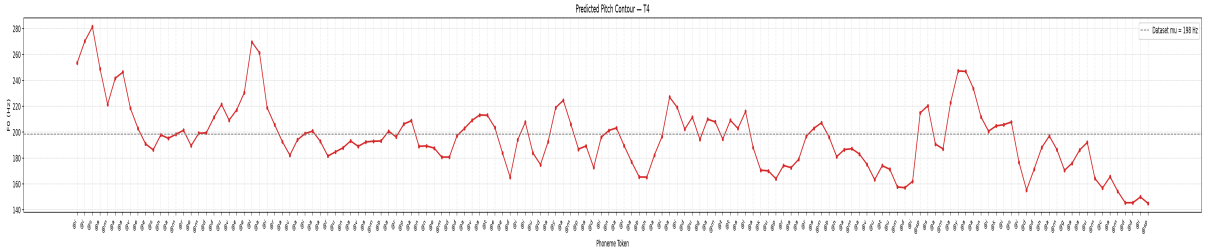


Figure 6.29: Predicted pitch contour for test sentence T4.

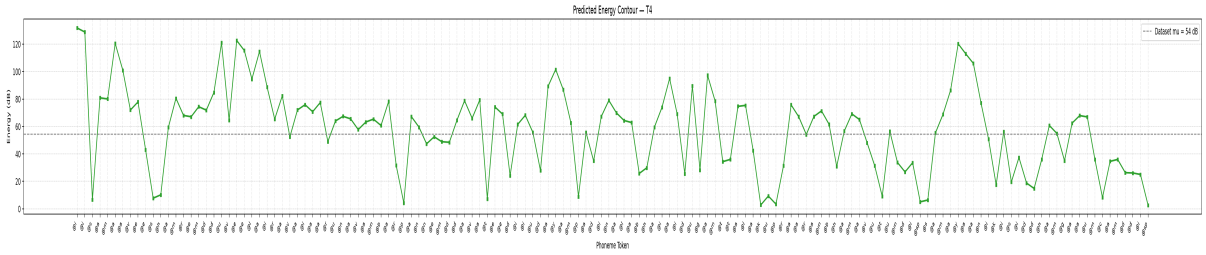


Figure 6.30: Predicted energy contour for test sentence T4.

7. Comparison: English vs. Telugu

7.1 Linguistic and Acoustic Differences

- **Rhythm.** English is stress-timed and reduces vowels in unstressed syllables (for example, the schwa in the verb form of *conduct*). Telugu is syllable-timed, with fairly even akshara durations. Lengthening occurs mainly on geminate consonants (పట్టు) and long vowels.
- **Tokenization.** English uses 100 IPA phoneme tokens and depends on a G2P lexicon. Telugu uses 460 akshara tokens taken directly from the native script, which avoids G2P out-of-vocabulary failures, although the model has to learn a larger set of symbol embeddings.
- **Recording consistency.** The Telugu data comes from clean studio recordings of two speakers, and its validation loss is lower (≈ 2.15) with a consistent voice. The English data was crowd-sourced from 104 speakers with different microphones and environments, and its validation loss is higher (≈ 2.59). Because the two systems differ across multiple confounding dimensions (corpus provenance, acoustic recording conditions, language family, token representation, and forced aligner architectures), the observed performance delta cannot be attributed to any single factor in isolation.

7.2 Limitations and Failure Cases

- **Word-boundary sound collision.** In standard synthesis, when long or vowel-starting words follow another word (such as *"The antidisestablishmentarianism"* or *"changed unexpectedly"*), the boundary sounds blur together. The end of the first word runs into the opening vowel, making the word onset unclear. In our tests, onset sound match dropped from 0.86 in isolation down to 0.72 in sentence context.
- **Short vowel dropping.** During fast speech, the duration model can assign only 0 or 1 frame (less than 16 ms) to weak vowels. This causes unstressed syllables to be swallowed or skipped.
- **Slowing down alone does not fix clarity.** In our experiments, simply stretching the whole sentence duration without boundary separation left the opening sound match unchanged at 0.729. This proved that speaking slower does not fix blurred word transitions.
- **Heteronym disambiguation.** FastSpeech 2 does not model sentence meaning. Distinguishing between noun and verb stress (such as *conduct*) depends entirely on the upstream text front-end.
- **High-frequency vocoder softening.** Resampling the vocoder from 22.05 kHz to 16 kHz slightly softens sharp consonant sounds like `/s/` and `/ʃ/`.

7.3 Tested Improvements and Future Work

1. **Vowel duration floor.** Enforcing a minimum length of 2 frames (32 ms) for all vowel sounds stops weak vowels from disappearing, keeping syllables complete without distorting the overall speaking rate.
2. **Selective boundary pauses.** Placing a trained micro-pause token (`@sp`) before difficult, vowel-starting words gives the vocal tract time to reset. In our experiments, this boosted onset clarity by +14 percentage points on *antidisestablishmentarianism* and +38 percentage points on *experiment*. Our testing also showed this must be selective: adding pauses before ordinary consonant words causes awkward hesitations.
3. **Multi-mode inference control.** Implementing selectable inference modes (standard baseline, duration-regulated, and combined pause mode) lets users choose between normal conversational rhythm and maximum phonetic clarity.
4. **Vocoder fine-tuning at 16 kHz.** Fine-tuning the HiFi-GAN vocoder directly on native 16 kHz audio would avoid resampling artifacts and sharpen unvoiced consonants.

8. Reproducibility

8.1 Prerequisites and Environment Setup

To reproduce this project from scratch, follow these setup steps in order:

8.1.1 Step 1: Clone the Repository

Clone the project repository from GitHub and navigate into the workspace directory:

```
git clone https://github.com/savioshaju/TTS.git
cd TTS
```

8.1.2 Step 2: Virtual Environment Creation

Use Python 3.11 to create and activate a virtual environment:

```
python -m venv .venv
.venv\Scripts\Activate.ps1
```

8.1.3 Step 3: Install Dependencies

Upgrade `pip` and install all pinned packages from `requirements.txt`:

```
python -m pip install --upgrade pip
pip install -r requirements.txt
```

The implementation runs on Python 3.11 with PyTorch 2.5, TorchAudio, NumPy, SciPy, Librosa, PyWorld, Matplotlib, and Transformers. Package versions are pinned in `requirements.txt`.

8.1.4 Step 4: Model Checkpoints and Vocoder Weights

- **HiFi-GAN vocoder.** Weights are located at:

  `English/inference/hifigan/generator_universal.pth.tar`
  `Telugu/inference/hifigan/generator_universal.pth.tar`

- **FastSpeech 2 checkpoints.** Model checkpoints are hosted on Google Drive:

  `https://drive.google.com/drive/folders/1MYS5zcsyehxDN5n4XaKj5jCidpUZ-efw?usp=sharing`

  Place the downloaded `best_model.pth` files into:

  English: `English/checkpoints/best_model.pth` (Step 61,000)
  Telugu: `Telugu/checkpoints/best_model.pth` (Step 74,000)

8.2 Dataset Download and Directory Setup

Both speech datasets are hosted on Hugging Face as sharded Parquet files that contain the audio, transcriptions, and speaker attributes:

- **English (AI4Bharat Svarah):** a multi-speaker, crowdsourced Indian English corpus (104 speakers across 19 native-language backgrounds). Download it with:

  ```
  pip install huggingface_hub
  huggingface-cli download ai4bharat/Svarah --repo-type dataset --local-dir
      datasets/Svarah
  ```

  The raw Parquet shards go in:

```
datasets/Svarah/data/test-*.parquet
```

- **Telugu (IndicTTS Telugu):** studio speech recordings curated by the Speech Technology Consortium at IIT Madras and MeitY (62.29 hours across 4 Parquet shards, 8,576 utterances from 1 female and 1 male studio speaker). Download it with:

```
huggingface-cli download SPRINGLab/IndicTTS_Telugu --repo-type dataset
    --local-dir datasets/IndicTTS_Telugu
```

The raw Parquet shards go in:

```
datasets/IndicTTS_Telugu/data/train-*.parquet
```

8.3 Preparing the 5-Hour Datasets

To meet the target of about 5 hours of training audio and 1 hour of validation audio per language, preparation scripts read the raw Parquet shards, filter by audio quality, sample in a stratified way, and export partitioned WAV files and manifests:

1. **English (`English/preprocess/create_svarah_splits.py`):**
   - Reads the Parquet shards in `datasets/Svarah/data/` and computes each clip's duration from its WAV header.
   - Identifies the 104 unique speakers from the demographic metadata (gender, age group, native state/district, mother tongue).
   - Keeps utterances in the TTS duration range ($1.0 \text{ s} \leq d \leq 12.0 \text{ s}$) and discards clipped audio.
   - Samples about 40 utterances per speaker across all 104 speakers, for a balanced acoustic representation (4,165 utterances, 6.01 hours in total).
   - Stages the extracted 16 kHz mono WAV files and pipe-delimited manifests to:

     ```
     datasets/prepared/english_speaker_independent/train/wavs/
     datasets/prepared/english_speaker_independent/train/metada
     ta.csv
     datasets/prepared/english_speaker_independent/val/wavs/
     datasets/prepared/english_speaker_independent/val/metadata
     .csv
     ```

     (Note: the final within-speaker stratified split of **3,419 training utterances (4.92 hours)** and **746 validation utterances (1.09 hours)** across all 104 speakers is established during feature extraction via `English/preprocess/repartition_and_renormalize_speaker_dependent.py` in Section 8.4).

```
python English/preprocess/create_svarah_splits.py
```

2. **Telugu (`Telugu/preprocess/create_telugu_splits.py`):**
   - Reads the Parquet shards in `datasets/IndicTTS_Telugu/data/` (8,576 utterances, 62.29 hours across 4 shards, from 1 female and 1 male studio speaker).
   - Keeps durations between $1.0 \text{ s} \leq d \leq 14.5 \text{ s}$ and requires a transcript length of at least 2.
   - Samples 5.00 hours in total with an exact 50/50 gender balance (2.50 hours female, 2.50 hours male), extracting 2,250 raw training utterances (5.0018 hours / 18,006 s) and 445 validation utterances (1.0030 hours / 3,611 s) with no overlap in utterances or transcripts (verified with SHA-256 text hashes). Following forced alignment (where 2 training files failed MMS boundary extraction), **2,248 aligned training utterances (4.999 hours / 17,996 s)** are used for model training and **445 validation utterances (1.003 hours / 3,611 s)** are held out for validation.

- Extracts the raw WAV files and metadata to:

  `datasets/prepared/telugu_prepared_v2/train/wavs/`
  `datasets/prepared/telugu_prepared_v2/train/metadata.csv`
  `datasets/prepared/telugu_prepared_v2/val/wavs/`
  `datasets/prepared/telugu_prepared_v2/val/metadata.csv`

```
python Telugu/preprocess/create_telugu_splits.py
```

8.4 Audio Standardization, Forced Alignment, and Feature Extraction

After the split, the raw audio is standardized, aligned, and converted into acoustic features:

1. **Audio standardization and resampling.**

   - English audio is checked to be 16 kHz mono PCM.
   - Telugu male audio (48 kHz, 1,433 files) is downsampled to 16 kHz with a 64-tap Kaiser-windowed polyphase sinc filter and an anti-aliasing cutoff of 7,600 Hz. Female studio audio (1,262 files) is already 16 kHz:

     ```
     python Telugu/preprocess/resample_telugu_audio.py
     ```

2. **Forced alignment.**

   - *English:* the Montreal Forced Aligner (MFA v2.2) aligns phonemes using the `english_mfa` acoustic model and pronunciation dictionary:

     ```
     python English/preprocess.py --config English/config.yaml --stage align
         --jobs 8
     ```

   - *Telugu:* Meta MMS (`facebook/mms-1b-all`) CTC forced alignment extracts frame-level boundaries for the native Telugu aksharas:

     ```
     python Telugu/preprocess/run_telugu_mms_alignment.py
     ```

3. **Acoustic feature extraction (`preprocessed_data`).** This step computes 80-channel log mel-spectrograms (STFT hop 256, window 1024), continuous F_0 contours with PyWorld Harvest, and short-time energy (L2-norm). The z-score statistics (μ, σ) are fitted on the training partition only and written to `preprocessed_data/stats.json`:

   ```
   python English/preprocess.py --config English/config.yaml --stage convert
   python Telugu/preprocess.py  --config Telugu/config.yaml
   ```

   Both pipelines write the standard FastSpeech 2 structure into `preprocessed_data/`:

   `preprocessed_data/train.txt`, `preprocessed_data/val.txt`
   `preprocessed_data/stats.json`, `preprocessed_data/speakers.json`
   `preprocessed_data/mel/`, `preprocessed_data/pitch/`
   `preprocessed_data/energy/`, `preprocessed_data/duration/`

4. **Speaker-dependent stratified repartitioning and feature re-normalization (English).** To ensure every one of the 104 Svarah speakers is represented in both training and validation subsets (enabling stable multi-speaker embedding optimization) and to strictly fit normalization statistics on the training subset, the speaker-dependent repartitioning script is executed on the extracted features:

   ```
   python English/preprocess/repartition_and_renormalize_speaker_dependent.py
   ```

This converts the initial speaker-independent extraction into the reported within-speaker stratified split of **3,419 training utterances (4.92 hours)** and **746 validation utterances (1.09 hours)**, writing the final `preprocessed_data/train.txt`, `preprocessed_data/val.txt`, and re-normalized `preprocessed_data/stats.json`.

8.5 Model Training: The Kaggle Dual-GPU Method

Both models were trained for 100,000 steps in Kaggle Notebooks with two NVIDIA Tesla T4 GPUs (**GPU T4 x2**, 16 GB VRAM each). The training procedure is packaged in two notebooks:

- English: `English/English_FastSpeech2_Kaggle.ipynb`
- Telugu: `Telugu/Telugu_FastSpeech2_Kaggle.ipynb`

Each notebook follows the same five stages:

1. **GPU and environment setup.** Detects the two T4 GPUs and initializes PyTorch `DataParallel` with AMP FP16. The batch size is 16 (8 utterances per GPU).
2. **Dataset extraction.** The staged archive (`preprocessed_data.zip`) is unpacked onto the local scratch disk for faster I/O.
3. **State saving.** Model checkpoints and optimizer states are saved to disk at regular intervals.
4. **Session resumption.** Kaggle limits a session to 12 hours, so training resumes from the latest saved checkpoint until all 100,000 steps are complete.
5. **Memory safeguards.** Periodic garbage collection and `malloc_trim(0)` help prevent memory fragmentation during multi-GPU runs.

8.6 Inference and Evaluation Commands

8.6.1 Speaker Selection and CLI Control

The English model conditions on the speaker through a learned embedding table (104×256), one entry for each of the 104 Svarah speakers. Speakers are indexed by integer IDs from 0 to 103 (mapped in `English/speakers.json`):

- **Default speaker (ID 27, `spk_029`).** Selected as the primary reference voice due to having the largest individual training utterance representation (118 utterances) in Svarah, ensuring natural prosody and stable phonetic alignment.
- **CLI parameter.** The voice can be set with the `--speaker_id` argument in both inference utilities (`inference.py` and `synthesize_custom.py`). If it is omitted, the default is Speaker ID 27 for English and Speaker ID 0 (`telugufemale`) for Telugu.

To synthesize all ten prescribed benchmark sentences (E1–E4 and T1–T4):

```
python English/generate_assignment_audios.py
python Telugu/generate_telugu_assignment_audios.py
```

8.6.2 Custom Synthesizer

To synthesize arbitrary input text with automatic rule-based text normalization:

```
python English/synthesize_custom.py \
    --text "The quick brown fox jumps over the lazy dog." \
    --output custom_synthesize/sample_custom.wav

python Telugu/synthesize_custom.py \
    --text "నమస్కారం, మీరు ఎలా ఉన్నారు?" \
    --output custom_synthesize/sample_telugu.wav
```

9. Conclusion

In this project, we built, trained, and evaluated two FastSpeech 2 text-to-speech systems: one for Indian English (using the Svarah dataset) and one for Telugu (using the IndicTTS dataset). The codebase was updated to train reliably on two GPUs, provide separate control over pitch and energy, and check text inputs carefully. For English, speech sounds were broken down into 100 phonetic units and aligned using Montreal Forced Aligner. For Telugu, speech was broken down into 460 akshara characters directly from Telugu script and aligned using Meta's MMS aligner.

Both models were trained for 100,000 steps on two NVIDIA Tesla T4 GPUs. While the validation loss flattened out early during training, this happened because human speech naturally has slight timing and pitch differences that loss math penalizes, not because the models stopped learning. Because of this, the best checkpoints were chosen by listening for clear speech and crisp consonants rather than just picking the lowest validation score. A pre-trained HiFi-GAN vocoder was used to turn predicted spectrograms into 16 kHz audio waveforms, with speed settings adjusted for natural speaking pace.

Testing on the ten assignment sentences showed clear and natural speech in both languages. The models spoke very long words, difficult consonant combinations, words with meaning-based stress (such as the noun and verb forms of *conduct*), dates, times, and words with accidental spaces. Overall, the project provides a clean FastSpeech 2 speech synthesis setup that is easy to run and reproduce for both English and Telugu.

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